

ML Model Implementation Report

Introduction

This project demonstrates the implementation of a machine learning classification model using the Titanic dataset. The objective of the project is to predict whether a passenger survived the Titanic disaster based on passenger information such as age, gender, passenger class, fare, and family details. Since the target variable, `survived`, has two possible values (0 = Did not survive and 1 = Survived), Logistic Regression was selected as the baseline classification model. The project covers the complete workflow, including data exploration, preprocessing, model training, and performance evaluation.

Dataset Description

The Titanic dataset is a publicly available dataset that is widely used for learning and practicing machine learning classification techniques. It contains information about passengers who traveled on the Titanic, along with their survival status. The dataset consists of **891 passenger records** and **12 features**.

Dataset Details

- **Dataset Name:** Titanic Dataset
- **Dataset Type:** Classification
- **Number of Records:** 891
- **Number of Features:** 12
- **Target Variable:** Survived

The dataset contains the following features:

- **PassengerId:** Unique identification number assigned to each passenger.
- **Survived:** Indicates whether the passenger survived (1) or did not survive (0).
- **Pclass:** Passenger class (1 = First Class, 2 = Second Class, 3 = Third Class).
- **Name:** Full name of the passenger.
- **Sex:** Gender of the passenger.
- **Age:** Age of the passenger in years.
- **SibSp:** Number of siblings or spouses traveling with the passenger.
- **Parch:** Number of parents or children traveling with the passenger.
- **Ticket:** Ticket number assigned to the passenger.
- **Fare:** Fare paid for the ticket.
- **Cabin:** Cabin number assigned to the passenger.
- **Embarked:** Port where the passenger boarded the ship (C = Cherbourg, Q = Queenstown, S = Southampton).

Exploratory Data Analysis (EDA)

The dataset was loaded into a Jupyter Notebook using the Pandas library. Basic exploratory data analysis was performed to understand the structure and quality of the data before building the machine learning model.

The following analyses were performed:

- Displayed the first five rows of the dataset using `head()`.
- Checked the shape of the dataset to determine the number of rows and columns.
- Examined the data types and non-null values using `info()`.
- Generated summary statistics for numerical columns using `describe()`.
- Checked for missing values using `isnull().sum()`.
- Verified that there were no duplicate records.

The analysis showed that the **Age**, **Cabin**, and **Embarked** columns contained missing values, while no duplicate records were found.

Feature Selection

Not all features in the dataset are useful for predicting passenger survival. Therefore, feature selection was performed to remove unnecessary columns and keep only the most relevant features.

The following columns were selected for model training:

- Pclass
- Sex
- Age
- SibSp
- Parch
- Fare
- Embarked

The following columns were removed because they were either unique identifiers, text-based information, or contained a large number of missing values:

- PassengerId
- Name
- Ticket
- Cabin

The **Survived** column was used as the target variable for the classification model.

Data Preprocessing

Before training the model, the dataset was preprocessed to handle missing values and convert categorical data into numerical form.

The following preprocessing steps were performed:

- Missing values in the **Age** column were filled using the median value.
- Missing values in the **Embarked** column were filled using the most frequent value (mode).
- The **Sex** column was encoded into numerical values (Male = 0, Female = 1).
- The **Embarked** column was encoded into numerical values (S = 0, C = 1, Q = 2).

These preprocessing steps ensured that the dataset was suitable for training the Logistic Regression model.

Train-Test Split

The processed dataset was divided into training and testing sets using Scikit-learn's `train_test_split()` function.

- **Training Data:** 80%
- **Testing Data:** 20%

The training dataset was used to train the machine learning model, while the testing dataset was used to evaluate its performance on unseen data.

Model Training

Since the target variable (**Survived**) contains two categories, Logistic Regression was selected as the baseline classification model.

The model was initialized with `max_iter=1000` to ensure that the optimization process had sufficient iterations to converge successfully. The model was then trained using the training dataset.

Model Evaluation

After the model was trained, predictions were made on the testing dataset. The performance of the model was evaluated using Accuracy and F1-score.

Evaluation Metric	Result
Accuracy	79.89%
F1-score	0.75

The Logistic Regression model achieved an **Accuracy of 79.89%** and an **F1-score of 0.75**. These results indicate that the model performs reasonably well in predicting passenger survival and provides a good baseline for this classification task.

Potential Improvements

Although the Logistic Regression model produced satisfactory results, the model can be further improved by:

- Performing additional feature engineering.
- Applying hyperparameter tuning to optimize model performance.
- Testing other machine learning algorithms such as Random Forest, Decision Tree, or Gradient Boosting.
- Using cross-validation to obtain a more reliable estimate of model performance.

Conclusion

This project successfully demonstrated the implementation of a machine learning classification model using the Titanic dataset. The workflow included loading the dataset, performing exploratory data analysis, selecting relevant features, preprocessing the data, splitting the dataset into training and testing sets, training a Logistic Regression model, and evaluating its performance. The model achieved an **Accuracy of 79.89%** and an **F1-score of 0.75**, showing that Logistic Regression is an effective baseline model for this binary classification problem. Overall, this project provided practical experience in applying the fundamental concepts of machine learning using Python and Scikit-learn.